

2.1

a. $\bar{x} = 5/5 = 1, \bar{y} = 10/5 = 2$

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

b. $b_2 = 8/10 = 0.8, b_1 = 2 - 0.8 * 1 = 1.2$

c. $\sum_{i=1}^5 x_i^2 = 3^2 + 2^2 + 1^2 + (-1)^2 + 0^2 = 15$

$$\sum_{i=1}^5 x_i y_i = 3 * 4 + 2 * 2 + 1 * 3 + (-1) * 1 = 18$$

$$\sum x_i^2 - N\bar{x}^2 = 15 - 5 * 1 = 10 \quad \therefore \sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$$

$$\sum x_i y_i - N\bar{x}\bar{y} = 18 - 5 * 1 * 2 = 8 \quad \therefore \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$$

d. $\hat{y}_i = 1.2 + 0.8x_i, \hat{e}_i = y_i - \hat{y}_i$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

$$s_y^2 = [2^2 + 1^2 + (-1)^2 + (-2)^2]/4 = 2.5$$

$$s_x^2 = [4 + 1 + 4 + 1]/4 = 10/4 = 2.5$$

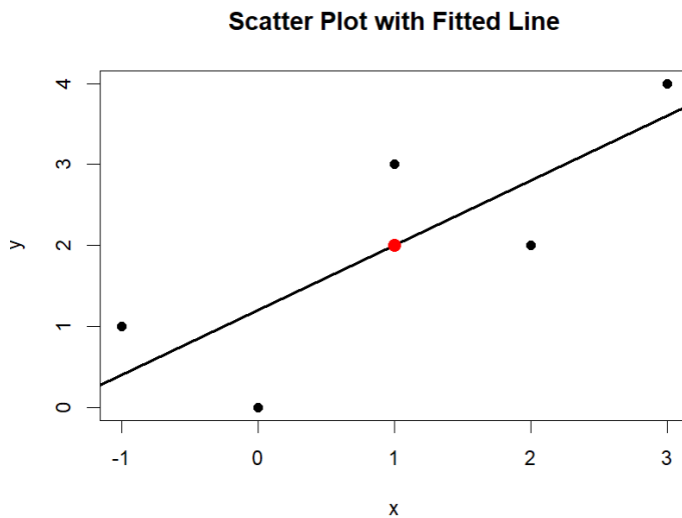
$$s_{xy} = (4 + 0 + 0 + 2 + 2)/4 = 2$$

$$r_{xy} = 2/\sqrt{2.5 * 2.5} = 0.8$$

$$CV_x = 100(\sqrt{2.5}/1) = 158.113883$$

Median of $x = 1$

e.



f. Yes. The fitted line passes through the point of the means, $\bar{x} = 1$, $\bar{y} = 2$.

g. $b_1 + b_2 \bar{x} = 1.2 + 0.8 * 1 = 2, \therefore \bar{y} = b_1 + b_2 \bar{x}$

h. $\bar{\hat{y}} = \sum \hat{y}_i / N = 10/5 = 2, \therefore \bar{\hat{y}} = \bar{y}$

i. $\hat{\sigma}^2 = \sum \hat{e}_i^2 / (N - 2) = 3.6 / (5 - 2) = 1.2$

j. $\widehat{var}(b_2 | x) = \hat{\sigma}^2 / \sum (x_i - \bar{x})^2 = 1.2 / 10 = 0.12$

$$se(b_2) = \sqrt{0.12} = 0.3464$$

$$22. a. TOTALSCORE = 916.442 + 12.175 * SMALL$$

The estimates are $\hat{\beta}_1 = 916.442$ and $\hat{\beta}_2 = 12.175$ ($p = 0.0236$). The intercept $\hat{\beta}_1$ is the mean TOTALSCORE for regular classes, and $\hat{\beta}_2$ is the mean difference between small and regular classes. Therefore, the regression suggests that being in a small class has a positive and statistically significant effect on learning, with the small-class students scoring about 12 points higher than regular-class students.

```
(Intercept)  916.442      3.675 249.396
small        12.175      5.369   2.268
              Pr(>|t|)
(Intercept)  <2e-16 ***
small        0.0236 *
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
  0.1 ' ' 1

Residual standard error: 74.59 on 773 degrees of freedom
Multiple R-squared:  0.006608, Adjusted R-squared:  0.005323
F-statistic: 5.142 on 1 and 773 DF, p-value: 0.02363
```

$$b. READSCORE = 432.665 + 6.924 * SMALL$$

The estimates are $\hat{\beta}_1 = 432.665$ and $\hat{\beta}_2 = 6.924$ ($p = 0.00151$). $\hat{\beta}_1$ is the mean reading score for regular classes and $\hat{\beta}_2$ is the mean difference in reading scores between small and regular classes. Since the result is statistically significant, there is strong evidence that small classes improve reading performance.

$$MATHSCORE = 483.777 + 5.251 * SMALL$$

The estimates are $\hat{\beta}_1 = 483.777$ and $\hat{\beta}_2 = 5.251$ ($p = 0.143$). $\hat{\beta}_1$ is the mean math score for regular classes and $\hat{\beta}_2$ is the mean difference in math scores. Although math scores are higher in small classes on average, the difference is not statistically significant.

This suggests that the small-class effect is positive and significant for reading, while the effect for math is positive but not statistically significant.

```
> summary(fit_b_read)

Call:
lm(formula = readscore ~ small, data = sub_a)

Residuals:
    Min       1Q   Median       3Q      Max
-74.665 -21.590  -2.665  16.410  194.335

Coefficients:
            Estimate Std. Error
(Intercept)  432.665      1.488
small         6.924       2.174
              t value Pr(>|t|)
(Intercept)  290.760 < 2e-16 ***
small        3.185  0.00151 **
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
  0.1 ' ' 1

Residual standard error: 30.2 on 773 degrees of freedom
Multiple R-squared:  0.01295, Adjusted R-squared:  0.01167
F-statistic: 10.14 on 1 and 773 DF, p-value: 0.001507
```

```
> summary(fit_b_math)

Call:
lm(formula = mathscore ~ small, data = sub_a)

Residuals:
    Min       1Q   Median       3Q      Max
-144.777  -35.028   -5.028    29.223   142.223

Coefficients:
            Estimate Std. Error
(Intercept)  483.777      2.449
small         5.251       3.578

            t value Pr(>|t|)
(Intercept) 197.545  <2e-16 ***
small         1.467   0.143
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*'
  0.05 '.' 0.1 ' ' 1

Residual standard error: 49.71 on 773 degrees of freedom
Multiple R-squared:  0.002778, Adjusted R-squared:  0.001488
F-statistic: 2.153 on 1 and 773 DF, p-value: 0.1427
```

$$c. TOTALSCORE = 916.442 + 4.306 * AIDE$$

The estimates are $\hat{\gamma}_1 = 916.442$ and $\hat{\gamma}_2 = 4.306$ ($p = 0.389$). The intercept $\hat{\gamma}_1$ is the mean TOTALSCORE for regular classes, and $\hat{\gamma}_2$ is the mean difference in TOTALSCORE (aide – regular). Since the p-value for $\hat{\gamma}_2$ is 0.389, the difference is not statistically significant. Therefore, there is no evidence that adding a teacher aide increases TOTALSCORE.

```
Call:
lm(formula = totalscore ~ aide, data = sub_c)

Residuals:
    Min       1Q   Median       3Q      Max
-191.748  -50.442   -5.442    40.558    312.558

Coefficients:
            Estimate Std. Error
(Intercept)  916.442      3.559
aide          4.306       4.994

            t value Pr(>|t|)
(Intercept) 257.529  <2e-16 ***
aide          0.862   0.389
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*'
  0.05 '.' 0.1 ' ' 1

Residual standard error: 72.23 on 835 degrees of freedom
Multiple R-squared:  0.0008898, Adjusted R-squared:  -0.0003068
F-statistic: 0.7436 on 1 and 835 DF, p-value: 0.3888
```

$$d. READSCORE = 432.665 + 2.871 * AIDE$$

The estimates are $\hat{\gamma}_1 = 432.665$ and $\hat{\gamma}_2 = 2.871$ ($p = 0.161$).

$$MATHSCORE = 432.665 + 2.871 * AIDE$$

The estimates are $\hat{\gamma}_1 = 483.777$ and $\hat{\gamma}_2 = 1.435$ ($p = 0.664$).

Both estimated aide effects are positive but not statistically significant. So, there is no evidence that adding a teacher aide improves achievement.

```
> summary(fit_d_read)
```

Call:

```
lm(formula = readscore ~ aide, data = sub_c)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-74.665	-21.536	-2.536	15.464	194.335

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	432.665	1.460	296.341	<2e-16 ***
aide	2.871	2.049	1.401	0.161

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 29.64 on 835 degrees of freedom

Multiple R-squared: 0.002347, Adjusted R-squared: 0.001152

F-statistic: 1.964 on 1 and 835 DF, p-value: 0.1615

```
> summary(fit_d_math)
```

Call:

```
lm(formula = mathscore ~ aide, data = sub_c)
```

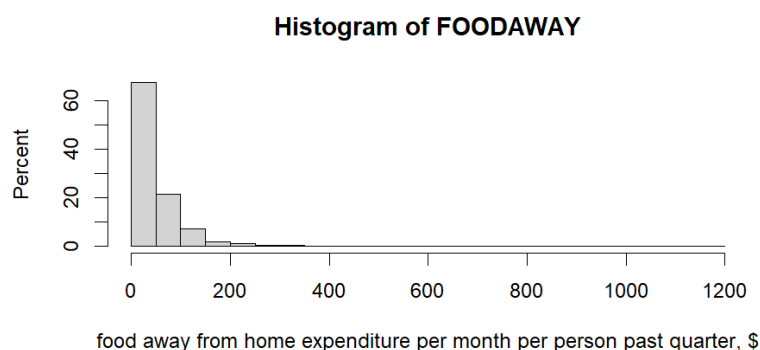
Residuals:

	Min	1Q	Median	3Q	Max
	-146.212	-31.212	-5.777	29.223	142.223

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	483.777	2.355	205.464	<2e-16 ***
aide	1.435	3.304	0.434	0.664

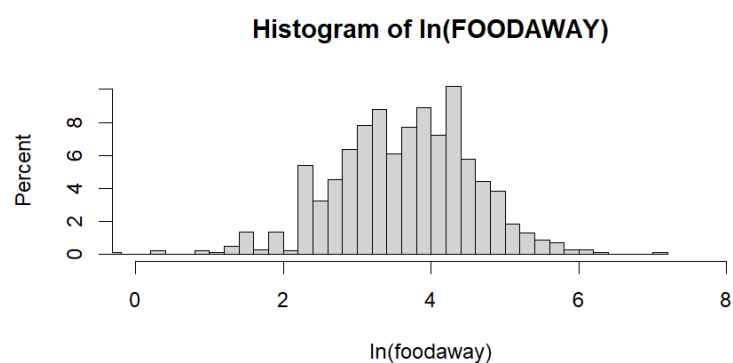
25. a. The mean is 49.27, median is 32.56, the 25th and 75th percentiles are 12.04 and 67.50.



b.

	mean	median
advanced	73.15	48.15
college	48.60	36.11
none	39.01	26.02

c.



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
-0.3011	3.0759	3.6865	3.6508	4.2797	7.0724	178

There are 178 fewer values of $\ln(\text{foodaway})$ because 178 households reported spending \$0 on food away from home per person, and $\ln(0)$ is undefined. These become missing values and are dropped from the regression.

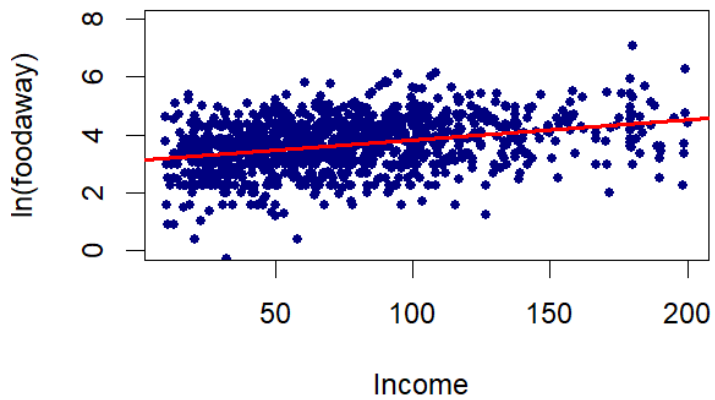
d. The estimated model is

$$\ln(\text{FOODAWAY}) = \beta_1 + \beta_2 \text{INCOME} + e = 3.1293 + 0.0069\text{INCOME}$$

Each additional \$100 household income results in an estimated 0.69% increase in food away expenditures per person, other factors held constant.

e. The plot shows a positive association between $\ln(\text{FOODAWAY})$ and INCOME s.

log-linear fitted line



f. The plot shows residuals randomly distributed with no obvious patterns. There are fewer observations at higher incomes, so there is more white space.

Residuals vs Income

